Then we get

$$AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = B.$$

Then $AX = B$ is a matrix equation involving matrices and it is called the **matrix form** of the system of linear equations (1). The matrix A is called the **coefficient matrix** of the system and the matrix $\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} & b_m \end{array}\right]$ is called the **augmented matrix** of the system. We denote the augmented matrix by $[A \mid B]$.

As an example, the matrix form of the system of linear equations

$2x + 3y - 5z + 7 = 0, 7y + 2z - 3x = 17, 6x - 3y - 8z + 24 = 0$ is $\begin{bmatrix} 2 & 3 & -5 \\ -3 & 7 & 2 \\ 6 & -3 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -7 \\ 17 \\ -24 \end{bmatrix}$.

1.4.3 Solution to a System of Linear equations

The meaning of solution to a system of linear equations can be understood by considering the following cases :

Case (i)

Consider the system of linear equations

$$2x - y = 5, \quad \ldots (1)$$
$$x + 3y = 6. \quad \ldots (2)$$

These two equations represent a pair of straight lines in two dimensional analytical geometry (see the Fig. 1.2). Using (1), we get

$$x = \frac{5 + y}{2}. \quad \ldots (3)$$

Substituting (3) in (2) and simplifying, we get $y = 1$.

Substituting $y = 1$ in (1) and simplifying, we get $x = 3$.

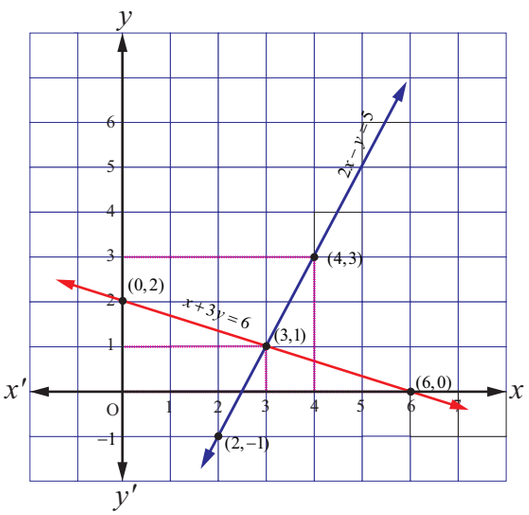


Fig.1.2

Both equations (1) and (2) are satisfied by $x = 3$ and $y = 1$.

That is, **a solution of (1) is also a solution of (2).**

So, we say that the system is consistent and has unique solution $(3,1)$.

The point $(3,1)$ is the point of intersection of the two lines $2x - y = 5$ and $x + 3y = 6$.

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